MA1522 Linear Algebra for Computing

Lesson 20: Range and Kernel of Linear Transformations

Announcements

- Today is the last lesson of new material.
- The remaining lessons are revision:
  - Fri 6 Nov: Chapters 1 & 2,
  - Tue 10 Nov: Chapters 3 & 4,
  - Fri 13 Nov: Chapters 5–6.

We have assignments for lesson 2 – 23. Among them 20 submission will be counted.

- **Final Exam**: **Tuesday 24 November 2026, 5:00 pm**. Closed book; one helpsheet is permitted. See the **Instruction of Final Exam** file on Canvas.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson20` (e.g. `A1234567Z_lesson20.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 20**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

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Section 7.1 Recall

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation.

- The **standard matrix** of T is $A = \begin{pmatrix} T(\mathbf{e}_1) & \cdots & T(\mathbf{e}_n) \end{pmatrix}$, and $T(\mathbf{u}) = A\mathbf{u}$ for all $\mathbf{u}$ in $\mathbb{R}^n$.
- If $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is a basis for $\mathbb{R}^n$, then

$$A = [T]_S \, P_{E \to S} = \begin{pmatrix} T(\mathbf{u}_1) & \cdots & T(\mathbf{u}_n) \end{pmatrix} \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{pmatrix}^{-1}.$$

- If $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$, then $T(\mathbf{v}) = c_1 T(\mathbf{v}_1) + \cdots + c_k T(\mathbf{v}_k)$. So the values of T on $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ determine T on their span.

Remark

Suppose that $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis of $\mathbb{R}^n$, and $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is an arbitrary set of vectors of $\mathbb{R}^m$. Then there exists a unique linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$T(\mathbf{v}_i) = \mathbf{u}_i \qquad \text{for } i = 1, \ldots, n.$$

pf

Uniqueness: Suppose T_1, T_2 are linear transformations from $\mathbb{R}^n \to \mathbb{R}^m$ s.t. $T_1(v_i) = u_i$, $T_2(v_i) = u_i$.

Then we show $T_1(x) = T_2(x) \quad \forall\, x \in \mathbb{R}^n$.

Since $\{v_1, \ldots, v_n\}$ is a basis, write $x = c_1 v_1 + \cdots + c_n v_n$.

$$T_1(x) = c_1 T_1(v_1) + \cdots + c_n T_1(v_n) = c_1 u_1 + \cdots + c_n u_n = c_1 T_2(v_1) + \cdots + c_n T_2(v_n) = T_2(x).$$

($T_1(v_1) = u_1$, $T_1(v_n) = u_n$)

Existence:

Consider $A = m\{(u_1 \cdots u_n)}_{n} \underbrace{(v_1 \cdots v_n)^{-1}}_{\text{a non square matrix}}$, $\quad [T]_S$

Then consider $T : \mathbb{R}^n \to \mathbb{R}^m$, $v \mapsto Av$

- $(u_1 \cdots u_n)(v_1 \cdots v_n)^{-1} \cdot v_i = u_i$.

($T(v_i)$, e_i)

Section 7.2 Range and Kernel

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation with standard matrix A.

Definition

- The **range** of T is $R(T) = \{ \mathbf{v} \in \mathbb{R}^m \mid \mathbf{v} = T(\mathbf{u}) \text{ for some } \mathbf{u} \in \mathbb{R}^n \}$.
- The **rank** of T is $\text{rank}(T) = \dim(R(T))$.
- The **kernel** of T is $\ker(T) = \{ \mathbf{u} \in \mathbb{R}^n \mid T(\mathbf{u}) = \mathbf{0} \}$.
- The **nullity** of T is $\text{nullity}(T) = \dim(\ker(T))$.

Exercise: a) $R(T)$ is always a subspace of $\mathbb{R}^m$.

b) $\ker(T)$ is always a subspace of $\mathbb{R}^n$.

Theorem

$R(T) = \text{Col}(A)$ is a subspace of $\mathbb{R}^m$, and $\ker(T) = \text{Null}(A)$ is a subspace of $\mathbb{R}^n$. Hence

$$\text{rank}(T) = \text{rank}(A), \qquad \text{nullity}(T) = \text{nullity}(A), \qquad \text{rank}(T) + \text{nullity}(T) = n.$$

Question 1

Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be the linear transformation

$$T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} x_1 + 2x_2 + x_3 + 3x_4 \\ -3x_1 - 6x_2 - 5x_3 - 11x_4 \\ -4x_1 - 8x_2 - 6x_3 - 14x_4 \end{pmatrix}.$$

1. Find a basis for $\ker(T)$, and write down $\text{nullity}(T)$.

2. Find a basis for $R(T)$, and write down $\text{rank}(T)$. Check that $\text{rank}(T) + \text{nullity}(T) = 4$.

Question 1 — solution: part 1

$$T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} x_1 + 2x_2 + x_3 + 3x_4 \\ -3x_1 - 6x_2 - 5x_3 - 11x_4 \\ -4x_1 - 8x_2 - 6x_3 - 14x_4 \end{pmatrix} \longrightarrow [T] = \begin{pmatrix} 1 & 2 & 1 & 3 \\ -3 & -6 & -5 & -11 \\ -4 & -8 & -6 & -14 \end{pmatrix}$$

Compute $R(T)$, $\ker(T)$.

$R(T) = \mathrm{Col}([T])$, $\ker(T) = \mathrm{Null}([T])$

ref

pivot columns

$$\begin{pmatrix} 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

- $\mathrm{Col}([T]) = R(T)$

$$= \mathrm{span}\left\{ \begin{pmatrix} 1 \\ -3 \\ -4 \end{pmatrix}, \begin{pmatrix} 1 \\ -5 \\ -6 \end{pmatrix} \right\} \Rightarrow \mathrm{rank}(T) = 2.$$

the columns in $[T]$ corresponding to pivot columns in ref.

- $\ker(T) = \mathrm{Null}([T])$

$$= \left\{ \begin{pmatrix} -2t-2s \\ t \\ -s \\ s \end{pmatrix} \;\middle|\; t, s \in \mathbb{R} \right\}$$

$$= \mathrm{span}\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ -1 \\ 1 \end{pmatrix} \right\} \Rightarrow \mathrm{Nullity}(T) = \mathrm{Nullity}(\overline{T}) = 2$$

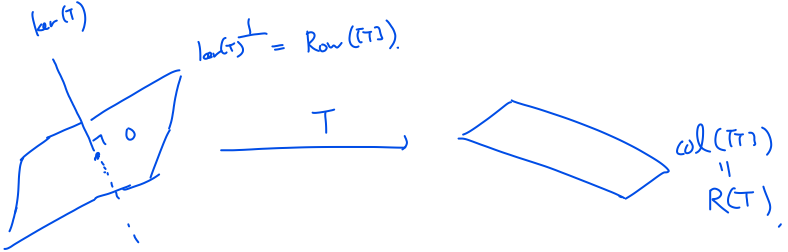
(t, s)

Question 1 — solution: part 2

Challenge 1

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation, and let $[T]$ denote its standard matrix.

Show that $\text{Row}([T])$ is the orthogonal complement of $\ker(T)$ in $\mathbb{R}^n$.



Section 7.2 Injective and Surjective

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation with standard matrix A.

general def for injectivity of functions.

Definition

T is **injective**, or one-to-one, if $T(\mathbf{u}_1) = T(\mathbf{u}_2)$ implies $\mathbf{u}_1 = \mathbf{u}_2$.

T is **surjective**, or onto, if for every $\mathbf{v}$ in $\mathbb{R}^m$ there is a $\mathbf{u}$ in $\mathbb{R}^n$ with $T(\mathbf{u}) = \mathbf{v}$, that is, $R(T) = \mathbb{R}^m$.

$\Rightarrow T(u_1) = T(u_2)$

$\Leftrightarrow T(u_1) - T(u_2) = 0$

$\Leftrightarrow T(u_1 - u_2) = 0$.

injectivity says: $T(u_1-u_2)=0 \iff u_1-u_2=0$.

Theorem

$\{v \mid T(v)=0\}$ = $\ker(T)$. $\quad$ Nullity$(T)=0$ $\quad$ Nullity(T) + rank(T) $= n$

- T is injective $\iff$ $\ker(T) = \{\mathbf{0}\}$ $\iff$ $\text{rank}(A) = n$.
- T is surjective $\iff$ $R(T) = \mathbb{R}^m$ $\iff$ $\text{rank}(A) = m$.
- If $m = n$: T is injective $\iff$ T is surjective $\iff$ A is invertible.

$Ax=0$ has only trivial sol

$Ax=b$ always have solution $\forall b$.

Question 2

True or false? Justify your answer.

(a) There is an injective linear transformation $T : \mathbb{R}^5 \to \mathbb{R}^3$.

(b) If $T : \mathbb{R}^3 \to \mathbb{R}^5$ is linear and the reduced row-echelon form of its standard matrix has 2 nonzero rows, then T is injective.

(c) If $T : \mathbb{R}^3 \to \mathbb{R}^3$ is linear and $T\begin{pmatrix}3\\0\\1\end{pmatrix} = T\begin{pmatrix}1\\4\\-2\end{pmatrix}$, then T is not surjective.

Question 2 — solution

a) There is an injective linear transformation $T: \mathbb{R}^5 \to \mathbb{R}^3$.

False.

$\text{Nullity}(T) + \text{rank}(T) = 5$.

$0 \le \dim(R(T)) \le 3$

$\{T(v) \mid v \in \mathbb{R}^5\}$.

$R(T) \subseteq \mathbb{R}^3$.

$\text{Nullity}(T) = 5 - \text{rank}(T) \ge 2 > 0 \implies \text{Nullity}(T) \neq 0 \implies T$ is not injective.

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b) If $T: \mathbb{R}^3 \to \mathbb{R}^5$, $\text{rref}([T])$ has 2 nonzero rows, then T is injective.

False.

$\text{rref}([T]) = \begin{pmatrix} * & ** \\ * & *** \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$ (3 columns) $\implies \text{rank}([T]) = \text{rank}(\text{rref}([T])) = 2$.

$\implies \text{Nullity}(T) = 3 - \text{rank}(T) = 1 \neq 0$.

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c) If $T: \mathbb{R}^3 \to \mathbb{R}^3$ linear, and $T\begin{pmatrix}3\\0\\1\end{pmatrix} = T\begin{pmatrix}1\\4\\2\end{pmatrix}$, then T is not surjective.

True.

T not injective
$\iff \ker(T) \neq 0 \iff \text{Nullity}(T) > 0$.
$\iff \text{rank}(T) = n - \text{Nullity}(T) < n$.

$\Downarrow$

T is NOT injective by thm.

Question 3

For a real number k, let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 3y - 2z \\ 2x + 7y - z \\ 4x + 11y + kz \end{pmatrix}.$$

1. Find all values of k for which T is injective. For the same k, is T surjective?

2. For the value of k where T is not injective, find a basis for $\ker(T)$, and find an equation in v_1, v_2, v_3 such that $\mathbf{v} = (v_1, v_2, v_3)^T$ is in $R(T)$ if and only if the equation holds.

Question 3 — solution: part 1

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 3y - 2z \\ 2x + 7y - z \\ 4x + 11y + kz \end{pmatrix}.$$

1. • Find all k st. T is injective. $\Leftrightarrow \operatorname{rank}([T]) = 3 \Leftrightarrow \det([T]) \neq 0 \Leftrightarrow k \neq -11$.

$$\det([T]) = \det\begin{pmatrix} 1 & 3 & -2 \\ 2 & 7 & -1 \\ 4 & 11 & k \end{pmatrix} = 1 \det\begin{pmatrix} 7 & -1 \\ 11 & k \end{pmatrix} - 3\det\begin{pmatrix} 2 & -1 \\ 4 & k \end{pmatrix} + (-2)\det\begin{pmatrix} 2 & 7 \\ 4 & 11 \end{pmatrix}$$

$$= 7k + 11 - 3(2k + 4) - 2 \cdot (-6)$$

$$= k + 11 \quad (= k + 11)$$

• When $k \neq -11$ T is inj $\Longrightarrow$ T is surj since $n = m = 3$.

Question 3 — solution: part 2

When $k = -11$. Find a basis for $\ker(T)$, and find an equation to describe $R(T)$.

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & x \\ 2 & 7 & -1 & y \\ 4 & 11 & -11 & z \end{array}\right)$$

Find condition on x, y, z st. this is consistent.

$\Updownarrow$

the constant column is not pivot

$\Updownarrow$

$-6x + y + z = 0$ ✱

$R_2 - 2R_1$
$R_3 - 4R_1$

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & x \\ 0 & 1 & 3 & y-2x \\ 0 & -1 & -3 & z-4x \end{array}\right) \xrightarrow{R_3 + R_2} \left(\begin{array}{ccc|c} 1 & 3 & -2 & x \\ 0 & 1 & 3 & y-2x \\ 0 & 0 & 0 & -6x+y+z \end{array}\right)$$

$$\text{Null}(T) = \left\{ \begin{pmatrix} 11t \\ -3t \\ t \end{pmatrix} \;\middle|\; t \in \mathbb{R} \right\} = \text{span}\left\{ \begin{pmatrix} 11 \\ -3 \\ 1 \end{pmatrix} \right\}$$

Challenge 2

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation, and let $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be the standard basis for $\mathbb{R}^n$. Let $V_i = \text{span}\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_i\}$, so we have a filtration

$$\{\mathbf{0}\} \subset V_1 \subset V_2 \subset \cdots \subset V_n = \mathbb{R}^n.$$

Show the following.

1. The standard matrix $[T]$ is upper triangular if and only if T preserves the subspaces V_i; namely, for every i and every $\mathbf{v} \in V_i$, $T(\mathbf{v}) \in V_i$.

2. The standard matrix $[T]$ is orthogonal if and only if for all $\mathbf{v}_1, \mathbf{v}_2$ in $\mathbb{R}^n$,

$$T(\mathbf{v}_1) \cdot T(\mathbf{v}_2) = \mathbf{v}_1 \cdot \mathbf{v}_2.$$

Next lesson

Next lesson (**Fri 6 Nov**), we begin the revision with

1. Chapter 1: Linear Systems, and
2. Chapter 2: Matrix Algebra.

Please bring the questions you would like to go through.